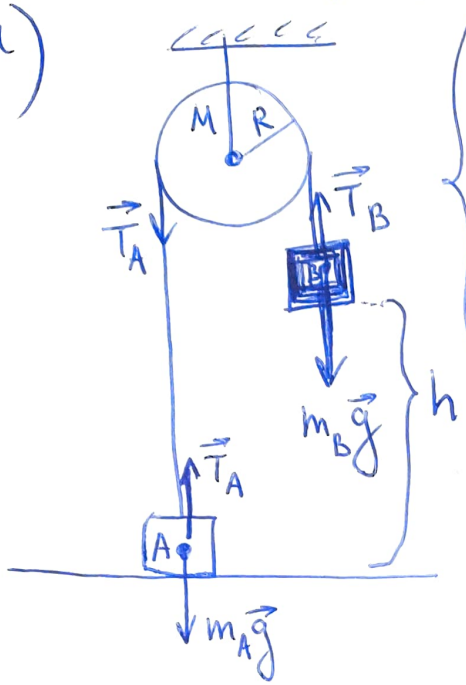
1 - masalaning yechimi. Ip chōzilmas (ideal) bōlgani uchun $|\vec{a}_A| = |\vec{a}_B| = a$

a)



$$\begin{cases} m_B g - T_B = m_B a & (1) \\ T_A - m_A g = m_A a & (2) \\ (T_B - T_A)R = \frac{1}{2} M R^2 \varepsilon & (3) \end{cases}$$

$$\varepsilon = \frac{a}{R}$$

(1) dan $T_B = m_B g - m_B a$ (4)

(2) dan $T_A = m_A a + m_A g$ (5)

(4) va (5) ni (3) ga qōyamiz

$$(m_B g - m_B a - m_A a - m_A g)R = \frac{1}{2} M R^2 \frac{a}{R}$$

$$m_B g - m_B a - m_A a - m_A g = \frac{1}{2} M a$$

$$(m_B - m_A) g = (m_A + m_B + \frac{1}{2} M) a$$

$$\boxed{a = \frac{m_B - m_A}{m_A + m_B + \frac{1}{2} M} \cdot g} \quad (6)$$

b) $h = \frac{at^2}{2} \Rightarrow$ $\boxed{t = \sqrt{\frac{2h}{a}} = \sqrt{\frac{2h}{g}} \cdot \sqrt{\frac{m_A + m_B + M/2}{m_B - m_A}}}$ (7)

c) $\boxed{v_B = a \cdot t = [(6) \cdot (7)] = \sqrt{2gh} \cdot \sqrt{\frac{m_B - m_A}{m_A + m_B + M/2}}}$

d) $\varepsilon = \frac{a}{R}$

$\varphi = \frac{\varepsilon t^2}{2} = \frac{at^2}{2R}$

$N = \frac{\varphi}{2\pi} = \frac{at^2}{4\pi R} = \frac{m_B - m_A}{2m_A + 2m_B + M} \cdot \frac{gt^2}{2R}$

$\boxed{N = \frac{h}{2\pi R}}$ Bu natijani ipning sirpanmasligi shartidan ham hosil qilsa bōladi.

2-masala yechimi

1-marta dam berilganda (porshen bosilganida)

$$P_0(V+V_0)^\gamma = P_1 V^\gamma$$

yani, $V+V_0$ hajm V gacha siqiladi, P_0 bosim P_1 gacha ortadi.

$$P_1 = P_0\left(\frac{V+V_0}{V}\right)^\gamma = P_0\left(1+\frac{V_0}{V}\right)^\gamma$$

2-marta dam berilishidan oldin nasos kamerasi havoga to'ldirilayotganda shlang klapani yopiladi va ballon ichidagi bosim P_1 qiymatni saqlaydi. Nasos kamerasi yana havoga to'lganida esa uning ichidagi P_0 atmosfera bosimini kla-panni ochishga yetarli bo'lmaydi. Porshen 2-marta bosilishni boshlaganidan toki nasos kamerasi ichidagi bosim P_1 ga tenglashmaguncha klapan ochilmaydi va kamera hajmi V_1 qiymatga erishganida bosim P_1 ga tenglashadi. V_1 ni topamiz:

$$P_0 V_0^\gamma = P_1 V_1^\gamma \qquad V_1 = \left(\frac{P_0}{P_1}\right)^{\frac{1}{\gamma}} \cdot V_0 = \frac{V_0}{1+\frac{V_0}{V}}$$

Klapan ochilganidan keyin esa 2-marta dam berishda ballonga havo kirib bosim P_2 ga teng bo'ladi, P_2 ni topamiz

$$\boxed{P_1(V_1+V)^\gamma = P_2 V^\gamma}$$

$$P_2 = P_1\left(1+\frac{V_1}{V}\right)^\gamma = P_1\left(1+\frac{V_0}{V+V_0}\right)^\gamma = P_0\left(1+\frac{V_0}{V}\right)^\gamma\left(1+\frac{V}{V+V_0}\right)^\gamma$$

$$P_2 = P_0\left(1+\frac{2V_0}{V}\right)^\gamma$$

Shunday qilib, bu mulohazalarni davom ettirib, P_n rekkurent munosabatni topamiz (matematik induksiya)

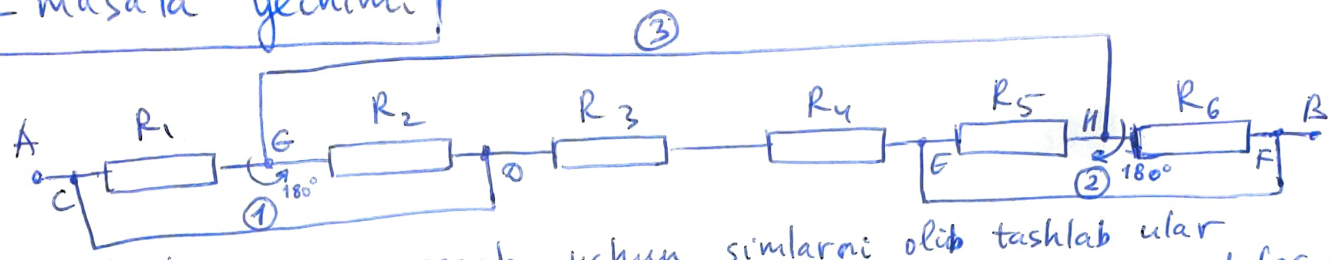
$$\boxed{P_n = P_0\left(1+n\frac{V_0}{V}\right)^\gamma}$$

$P_n = NP_0$ shartdan n ni topamiz:

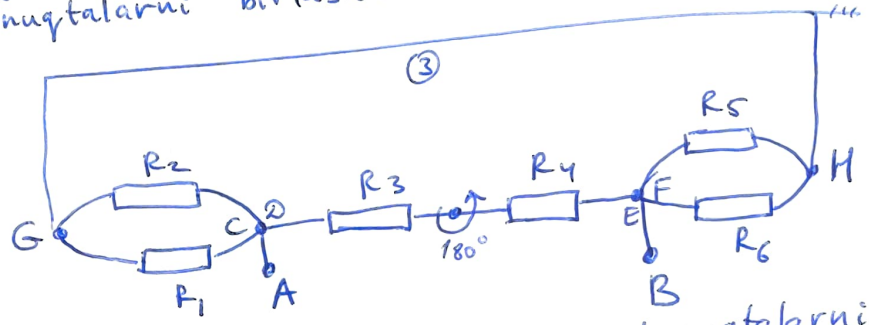
$$NP_0 = P_0\left(1+n\frac{V_0}{V}\right)^\gamma$$

$$\boxed{n = \frac{V}{V_0}\left(N^{\frac{1}{\gamma}}-1\right)}$$

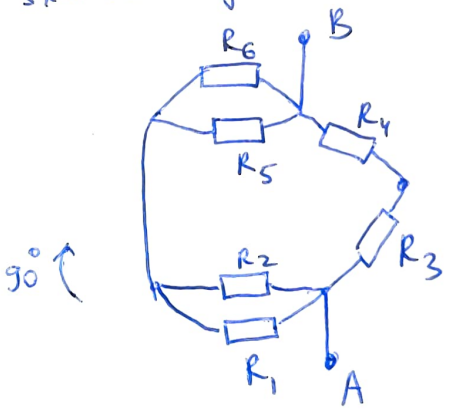
3- masala yechimi

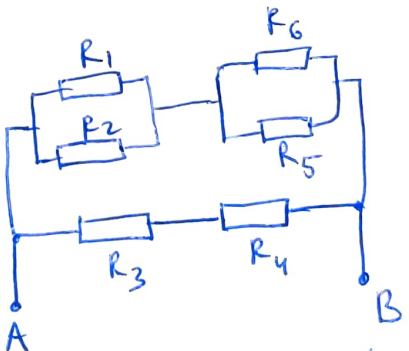


Ekvivalent sxema yasash uchun simlarni olib tashlab ular tutashtiradigan nuqtalarni birlashtirish uchun sxemani deformatsiyalaymiz. Oldin ① va ② simlarni uzib C va D hamda E va F nuqtalarni birlashtiramiz.



Keyin esa ③ simni uzib G va H nuqtalarni birlashtirish uchun sxemani yana deformatsiyalaymiz.





Ekvivalent sxema hosil bo'ldi. R_1 va R_2, R_5 va R_6 parallel.

$$R_{12} = \frac{R_1 R_2}{R_1 + R_2} = 0{,}5\ \Omega \qquad R_{56} = \frac{R_5 \cdot R_6}{R_5 + R_6} = 0{,}5\ \Omega$$

R_3 va R_4 ketma-ket ulangan $R_{34} = R_3 + R_4 = 2\ \Omega$

$$R_{AB} = \frac{R_{34} \cdot (R_{12} + R_{56})}{R_{34} + R_{12} + R_{56}} = \frac{2 \cdot (0{,}5 + 0{,}5)}{2 + 0{,}5 + 0{,}5} = \frac{2}{3} \approx 0{,}66\ \Omega$$

4-masala yechimi

$$\mathcal{E}_n = -\frac{d\Phi}{dt} = -\frac{d(BS)}{dt} = -S\frac{dB}{dt} = -S B_0 \omega \cos(\omega t)$$

$$\mathcal{E}_n = -\pi R_n^2 B_0 \omega \cos(\omega t) \qquad R_n = n\Delta$$

$$\mathcal{E}_n = -n^2 \pi \Delta^2 B_0 \omega \cos(\omega t) \qquad \boxed{N = \frac{R}{\Delta}}$$

$$\mathcal{E} = \sum_{n=1}^{N} \mathcal{E}_n = -\pi \Delta^2 B_0 \omega \cos(\omega t) \sum_{n=1}^{N} n^2$$

5-masala yechimi

Fotonning massasi bu modelda $m = \frac{h\nu}{c^2}$ formula bilan aniqlanadi. Foton yulduzdan chiqib, o'zining energiyasini yo'qotadi. Bu yo'qotilgan energiya gravitatsion maydon tomonidan bajarilgan ishga teng:

$$dE = -F\,dr$$ (kichik dr masofaga siljiganda yo'qotilgan energiya dE). Minus ishora F va c lar qarama qarshi yo'nalgani uchun, yoki energiya kamaygani uchun olinadi

$$d(h\nu) = -\frac{\gamma m M}{r^2}\,dr$$

$$h\,d\nu = -\frac{\gamma M}{r^2}\frac{h\nu}{c^2}\,dr$$

$$\int_{\nu_1}^{\nu_2} \frac{d\nu}{\nu} = -\frac{\gamma M}{c^2}\int_{R}^{\infty} \frac{dr}{r^2}$$

$$\ln\left(\frac{\nu_2}{\nu_1}\right) = \frac{\gamma M}{c^2 R}$$

$$\frac{\nu_2}{\nu_1} = e^{\frac{\gamma M}{c^2 R}}$$

$$\frac{\nu_1 - \Delta\nu}{\nu_1} = e^{\frac{\gamma M}{c^2 R}}$$

$$1 - \frac{\Delta\nu}{\nu_1} = e^{\frac{\gamma M}{c^2 R}}$$

$$\frac{\Delta\nu}{\nu_1} = 1 - e^{\frac{\gamma M}{c^2 R}}$$

$$e^x \approx 1 + x \qquad x \ll 1$$

$$\frac{\Delta\nu}{\nu_1} = 1 - 1 - \frac{\gamma M}{c^2 R}$$

$$\frac{\Delta\nu}{\nu_1} = -\frac{\gamma M}{c^2 R}$$

$$\boxed{\frac{M}{R} = -\frac{c^2}{\gamma}\frac{\Delta f}{f}}$$

$\Delta f < 0$, $\frac{M}{R} > 0$